

GUAN JIE KHAMCHAN

Full-stack Developer · React · TypeScript · Node.js

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Available for a 5-month cooperative education placement · January – May 2027 · Bangkok | Work authorisation: Thailand and Malaysia

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Java, Python, SQL
Frontend	React, Next.js, Vue.js, TypeScript, Tailwind CSS, Radix UI, react-hook-form, HTML5, CSS3
Backend	Node.js, Express, Spring Boot, Hibernate, REST API design, OpenAPI / Swagger
Databases	MongoDB, Redis, MySQL
DevOps	Docker, Docker Compose, Nginx, GitHub Actions, GitLab CI
ML & CV	PyTorch, OpenCV, Albumentations, NumPy

PROJECTS

Land Cover Semantic Segmentation — Deep learning pipeline Aug 2025 – Present

[PyTorch](#) · [EfficientNet-B4](#) · [OpenCV](#) · [NumPy](#) — github.com/Leng201202/lulc_cei

- Owned the pipeline end to end — dataset loading, model and loss construction, training loop, validation, checkpointing, evaluation and inference — for land-use / land-cover segmentation on the **OpenEarthMap** dataset.
- Built a **U-Net with an EfficientNet-B4 encoder** producing 8-class pixel predictions, trained with a **combined CrossEntropy + Dice loss** and AdamW so rare classes were not lost to the dominant background.
- Implemented **mixed-precision training** to halve GPU memory use, plus full-image and **tiled inference** so large scenes could be predicted without downsampling away small features.
- Built the evaluation harness reporting **overall accuracy, mIoU and mean F1**, with fine-tuning from pretrained weights to shorten experiment cycles.

Digital University E-Questionnaires Platform — Full-stack survey system Nov 2025 – Aug 2026

[Vue.js](#) · [Node.js](#) · [Express](#) · [MongoDB](#) · [Redis](#) · [Docker](#) · [Nginx](#) — github.com/Napus-BackendDev/Digital-University-Project-SE

- Built an enterprise questionnaire platform for educational institutions on a **decoupled Vue.js + Node/Express architecture**, containerised end to end with **Docker Compose and Nginx** so the full stack deploys with a single command — **448 commits** across the team.
- Built the **dynamic form-builder interface** that lets administrators compose questionnaires without developer involvement, and implemented the **API integration layer** wiring the Vue front end to the Express backend across the platform.
- Contributed to **role-based access control** and a real-time analytics dashboard built with Chart.js, backed by **MongoDB 7.0** with **Redis** caching on read-heavy report queries.
- Documented the API with **Swagger / OpenAPI 3.0** and added a developer role-switcher that removed the manual re-login loop during testing.

Dormitory Booking System — Frontend owner Aug 2025 – Nov 2025

[Next.js 14](#) · [React 18](#) · [TypeScript](#) · [Tailwind CSS](#) · [Radix UI](#) — github.com/Napus-BackendDev/Dormitory-Booking-System

- Owned the entire frontend** of a dormitory booking and maintenance platform, built with **Next.js 14 and React 18 in TypeScript**, styled with **Tailwind CSS** and composed from **Radix UI** primitives so components were keyboard-accessible by default.
- Implemented the booking and request flows with **react-hook-form** for client-side validation, plus **Recharts** dashboards and toast notifications, keeping every screen responsive across mobile and desktop.
- Integrated the UI against the team's **OpenAPI specification**, letting the front end be built in parallel with the backend rather than waiting on it.

EDUCATION

Mae Fah Luang University — Bachelor of Engineering, Software Engineering Expected 2027

Relevant coursework: Software Engineering, Web Application Development, Database Systems, Data Structures & Algorithms, Machine Learning

ACTIVITIES & ADDITIONAL

- MFU Software Engineer Hackathon 2025** — built a mobile sleep-tracking application in the health track with a team of four, taking the idea from concept to a working prototype within the event.
- Languages:** Mandarin Chinese (native — spoken and written), Thai (fluent), English (intermediate working proficiency)